Ryan Peckham

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Education

University of Michigan — Ann Arbor, MI

B.S. Computer Engineering; GPA: 3.35/4.0 — Aug. 2025 – Present
Sophomore – Anticipated Graduation May 2029
Coursework: Multivariable & Vector Calculus, Electricity & Magnetism, Quantum Photonics, Discrete Mathematics, Solar Energy Systems, Probability & Statistics

Skills

Programming Languages: Python, Rust, C, C++, C#, Java, SQL, MATLAB
Web: HTML, CSS, JavaScript, Svelte, Cloudflare Workers
Systems: Git & GitHub, PostgreSQL, WSL, Docker, Database Normalization, Threading
Operating Systems: Windows, Linux (Ubuntu, Debian, NixOS), MacOS
Hardware: Soldering, Shielded Metal Arc Welding, CNC Machines, Lathes, Mills

Experience

Michigan Autonomous Security Team (M-AST), Chief Software Engineer — Jan. 2026 – Present
University of Michigan
- Leading and mentoring a team of engineers to install, program, and test gyroscope, magnetometer, barometer, and GPS sensors onto a prototype drone
- Maintaining the organization of a large and complex codebase and reviewing code and pull requests
- Co-designed the drone's system architecture with the M-AST President, defining data flow between subsystems and supporting software-in-the-loop (SITL) and hardware-in-the-loop (HITL) runtimes
- Integrating ROS2, OpenCV, telemetry, and path-finding modules into a cohesive flight software stack

Michigan Hackers, Machine Learning Team, Member — Nov. 2025 – Present
University of Michigan
- Built a foundation in machine learning theory including Markov chains, Bayes' theorem, backpropagation, regression, and neural network design
- Led peer lectures and contributed to technical discussion of modern theory and applications of machine learning

Engineering 100 — Aug. 2025 – Dec. 2025
University of Michigan
- Built a sun-tracking solar panel on a Raspberry Pi, writing the C++ control code and wiring and soldering photoresistors, stepper motors, voltage limiters, and a battery
- Collaborated in a design team to prioritize tasks, balance planning with execution, and draw meaningful conclusions from test data

FIRST Robotics Competition (FRC), Team 141 "WOBOT" — Aug. 2023 – Oct. 2024
West Ottawa High School, Programming Team
- Developed input and movement code by implementing and tuning PIDF loops for precise control
- Contributed to a District Championship-winning team that also earned the Engineering Inspiration, Sustainability, and Entrepreneurship awards

Cultural, Technological, and Environmental Exchange (CTEE) Club — Sep. 2024 – Apr. 2025
West Ottawa High School
- Diagnosed and refurbished desktop systems, troubleshooting hardware-level faults on motherboards, power supplies, and storage drives
- Replaced faulty physical components and managed hardware integration, deepening practical understanding of computer architecture
- Provisioned hardware by deploying Microsoft Windows OS and configuring system drivers, preparing ready-to-use systems for shipment to Cambodia and Haiti

Personal Projects

- Designed, built, and deployed my personal website peckhamr.com with Svelte, JavaScript, HTML, and CSS, teaching modern myself web design and maintainable codebase design
- Designed and built M-AST's public website, flymast.com, delivering a polished, professional presence for prospective students and sponsors

Work Experience

Ace Hardware, Delivery and Assembly Technician — May 2026 – Aug. 2026
Chick-fil-A, Crew Member — June 2025 – Jan. 2026
Target, Electronics Consultant & Team Leader — Nov. 2022 – Aug. 2024